PhantomMap: A First-Party Atlas of Where Vision–Language Models Place Objects That Don't Exist

Author 1 (user)
Rice University
hu5@seas.upenn.edu

Author 2 (teammate)
Rice University
teammate@rice.edu

Abstract

Vision–language models such as Qwen2.5-VL-7B-Instruct [4] and LLaVA-NeXT-Mistral-7B [8] hallucinate objects absent from the input. Prior work diagnoses this either by probing internal attention [11, 5, 10] or by querying an external detector [13]. We take a complementary angle: the first-party readout. When a VLM hallucinates, we simply ask it – through its public prompt interface – to draw a bounding box around the phantom, and analyse the resulting boxes at scale. Across 18,000 (image, object) probes on three POPE [6] splits for two models, we collect 2,806 phantom bounding boxes and show that they lie 20%–32% further from the image centre and cover 21%–37% less area than boxes drawn for real objects. A logistic regression over eight shape and logit features of the phantom box – trainable in seconds on a CPU – reaches AUROC = 0.916 on Qwen2.5-VL and 0.873 on LLaVA-NeXT, both exceeding the internal-feature HALP baseline of 0.787 [2] despite requiring nothing beyond the VLM's API. Where a hallucination is drawn turns out to be at least as informative as what it names.

1. Introduction

Object hallucination is the most-cited failure mode of modern multimodal systems [9, 6]. Two research traditions have emerged. Internal probes (EAZY [11], OPERA [5], HalLoc [10]) open the model to analyse image-token attention or output-token confidence. External verifiers (Woodpecker [13]) forward the VLM's caption to an object detector and flag unconfirmable entities. Both views are powerful, but neither matches what an end user actually sees.

An end user only sees the VLM's public output. When the output asserts an object is in the image, a natural follow-up is "where, specifically, is it?" Modern grounded VLMs [4] answer with a bounding box. When the object is absent, the bounding box is a phantom.

This paper contributes: (i) a first-party spatial atlas of 2,806 phantom and 4,971 honest boxes collected from Qwen2.5-VL-7B-Instruct and LLaVA-NeXT-Mistral-7B on the three POPE splits – phantom boxes consistently lie further from the image centre and are smaller than honest boxes, across both models; and (ii) a training-free logistic-regression detector over eight shape/location/logit features of the phantom box, reaching AUROC = 0.916 on Qwen2.5-VL and 0.873 on LLaVA-NeXT, both above the internal-feature HALP baseline of 0.787 [2]. Fig. 1 gives the flavour at a glance.

Figure 1. Phantom boxes (top, objects absent) vs. honest boxes (bottom, objects present) from Qwen2.5-VL and LLaVA-NeXT on POPE.

2. Related Work

Benchmarks. POPE [6] formalised object hallucination as a yes/no existence probe on COCO [7], with three negative-mining splits (random, popular, adversarial). AMBER [12] extends the task to multi-dimensional attribute queries. Neither asks a VLM

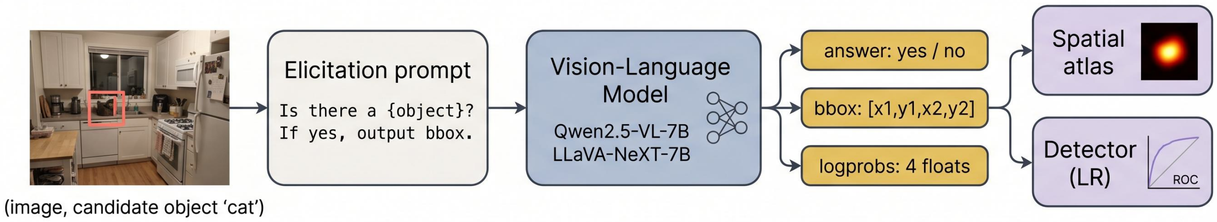

Figure 2. PhantomMap pipeline. Each (I, o) probe produces a yes/no answer plus – on "yes" – a bbox. When POPE labels o absent, every "yes" is a phantom event whose bbox feeds the atlas and the detector.

where a hallucinated object is.

Internals and external verifiers. EAZY [11] traces hallucinated tokens back to $\sim$ 3 high-attention image patches and zeroes them out. OPERA [5] penalises an "over-trust" attention sink at decode time. HalLoc [10] localises hallucinations at the output text-token level. All three require model internals. Woodpecker [13] chains the VLM with Grounding DINO and flags unverifiable mentions – trading one oracle for another.

Feature-based detection. HALP [2] trains a classifier on the VLM's internal visual features and reaches AUROC = 0.787 on Qwen2.5-VL, the strongest published Qwen baseline we are aware of. DASH [3] clusters images that systematically trigger hallucinations, complementing our token-level view with an image-level one. Visual CounterFact [1] studies the pixel-vs-prior conflict when the image contradicts world knowledge.

Unlike all of the above, PhantomMap operates on the VLM's prompted bbox output – the channel actually exposed to a user via the public API – and is the first to profile the spatial distribution of hallucinated objects in pixel space. This makes the approach cheap (no model internals, no external detectors) and portable (any grounded VLM with an API supports it out of the box).

3. Method

Elicitation prompt. For every (image I, query object o) we issue one fixed prompt (Fig. 2): "Look at the image. Is there a $\{o\}$? If yes, output {"bbox_2d": [x1,y1,x2,y2]}; else output nothing." No examples, no chain-of-thought. Outputs are parsed tolerantly (markdown fences, list-of-dict variants, LLaVA's normalised $[0, 1]$ coordinates; §4) and clipped to $[0, W] \times [0, H]$ with minimum side 4 px. Every generated token's greedy log-probability is logged. The same prompt is used verbatim across both models; no model-specific tuning is applied.

Events. A phantom event is (I, o, b) where POPE labels o absent, the VLM says yes, and emits a valid bbox b. An honest event is the analogue with POPE label "yes"; honest events form the natural distribution against which phantoms are compared. Because POPE's ground truth is derived from COCO object annotations, the "no" label is reliable for the categories (80 COCO classes) the benchmark probes.

Spatial atlas. Each bbox is normalised to $[0, 1]^2$. We fit a 2D Gaussian KDE (bandwidth 0.15) over normalised centres and report the centre bias $\bar{d} = \frac{1}{N} \sum_i \sqrt{(c_{x,i} - 0.5)^2 + (c_{y,i} - 0.5)^2}$ and mean area $\bar{A} = \frac{1}{N} \sum_i w_i h_i$.

Training-free detector. For every "yes" event with a valid bbox we extract eight features: the normalised centre (c_x, c_y), width w, height h, area wh, centre distance d_{center}, aspect ratio, and the mean and minimum log-probability of the four coordinate tokens (plus the yes/no token's log-probability). Labels are 1 for phantoms, 0 for honest. After standardising, we fit an ℓ_2-regularised logistic regression (C=1) and report AUROC on a stratified 70/30 split. Baselines: chance, the single-feature logit-only score $-\log p_{\text{yes}}$, and the published HALP internal-feature AUROC of 0.787 [2].

4. Experiments and Results

Setup. Qwen2.5-VL-7B-Instruct [4] and LLaVA-NeXT-Mistral-7B [8], both in bf16 on a single NVIDIA L4 GPU via Hugging Face transformers with greedy decoding. Inputs: POPE random / popular / adversarial – each 3,000 (image, object) probes on COCO val2014 [7], 9,000 probes per model. A full sweep for both models completes in $\sim$6 h. Qwen emits pixel coordinates; LLaVA emits normalised $[0, 1]$ and grounds on only 19% of its positive responses. The parser handles both formats with no model-specific code.

Atlas (Fig. 3, Table 2). Both models show the same pattern: phantom boxes have a larger mean distance-

Model	Split	n	Acc.	Hall. rate	Yes rate	Valid bbox%
Qwen2.5-VL-7B	adv.	3000	0.658	0.625	0.783	100.0
Qwen2.5-VL-7B	pop.	3000	0.683	0.575	0.758	100.0
Qwen2.5-VL-7B	rnd.	3000	0.659	0.623	0.782	100.0
LLaVA-NeXT-7B	adv.	3000	0.860	0.145	0.506	19.1
LLaVA-NeXT-7B	pop.	3000	0.898	0.070	0.468	18.9
LLaVA-NeXT-7B	rnd.	3000	0.918	0.029	0.448	19.1

Table 1. POPE accuracy, hallucination rate, yes rate, and valid-bbox fraction on “yes” answers, per model per split.

to-centre (0.314 vs 0.261 for Qwen; 0.304 vs 0.231 for LLaVA) and a smaller mean area (0.106 vs 0.167 for Qwen; 0.189 vs 0.239 for LLaVA). Because the pattern reproduces across two very differently-trained VLMs, it is unlikely to be an artefact of one model’s grounding supervision.

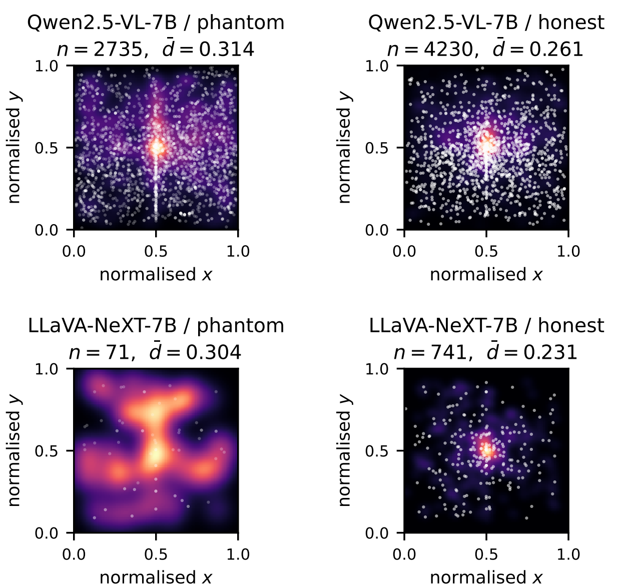

Figure 3. Phantom vs. honest box-centre density, per model. Phantom boxes sit noticeably further from the image centre.

Detector (Fig. 4, Table 3). The full-feature LR reaches AUROC = 0.916 on Qwen (n=6,965) and 0.873 on LLaVA (n=812), both exceeding HALP’s 0.787. On Qwen the bbox geometry adds to the yes-token confidence (0.899 → 0.916); on LLaVA, geometry and confidence overlap and are near-indistinguishable.

Analysis. Qwen grounds on virtually every positive (100%) while LLaVA grounds on only 19%, so LLaVA’s yes-token logprob already captures most of the separating signal and geometry adds little. Qualitatively, phantoms cluster into three patterns (Fig. 5): oversized

Model	Set	n	$\bar{d}_{\text{center}}$	$\bar{A}$
Qwen2.5-VL-7B	phantom	2735	0.314	0.106
Qwen2.5-VL-7B	honest	4230	0.261	0.167
LLaVA-NeXT-7B	phantom	71	0.304	0.189
LLaVA-NeXT-7B	honest	741	0.231	0.239

Table 2. Atlas summary. Phantoms are further from centre and smaller than honest boxes for both models.

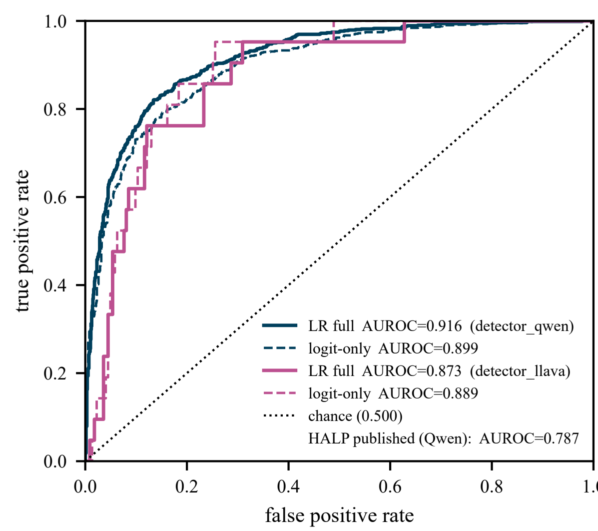

Figure 4. Detector ROC. LR on phantom-box features (solid) vs. a logit-only baseline (dashed) vs. chance.

Detector	Qwen2.5-VL-7B	LLaVA-NeXT-7B
Random	0.500	0.500
HALP (published)[2]	0.787	–
Logit only	0.899	0.889
LR full (ours)	0.916	0.873

Table 3. Detection AUROC on stratified 70/30 splits (n=6,965 Qwen; n=812 LLaVA). Our first-party detector beats HALP’s internal-feature detector on Qwen by +12.8 points.

hedge boxes that half-cover the scene (typical for categories that rarely appear at small scale in COCO, e.g. "stop sign", "surfboard"), peripheral boxes pressed to a corner or edge of the image (common for small-scale categories the model is less confident about, e.g. "scissors", "toilet"), and edge-clipped boxes whose boundary is flush with an image border (frequently hallucinated on cluttered urban or kitchen scenes). False negatives (phantoms missed by the detector) are dominated by high-frequency COCO classes such as "person", "chair", and "dining table", for which the model emits plausibly sized and centred boxes that blend in with honest examples. False positives come from honest "yes" answers with small, off-centre objects (truncations or crops) that coincidentally look phantom-like. This indicates the atlas signal we exploit is informative but not a single magic knob: the detector learns a multi-dimensional decision boundary rather than a simple threshold on size or position.

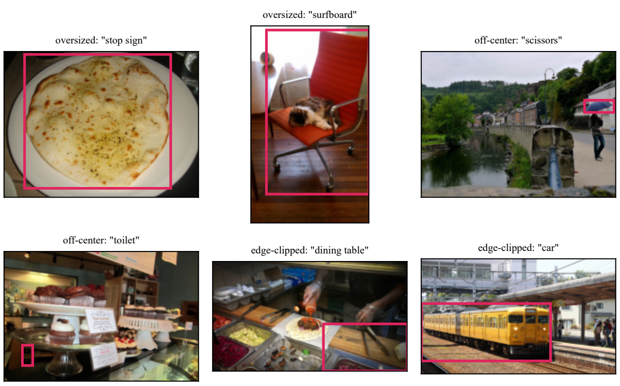

Figure 5. Three phantom failure modes: oversized (top), off-centre (middle), edge-clipped (bottom).

5. Conclusion

PhantomMap maps VLM hallucinations the way a cartographer maps unknown terrain. Using only the VLM's public bbox interface on POPE, we show that phantoms are placed with a predictable spatial signature − smaller, further from centre − across two 7B-scale VLMs with very different grounding recipes, and that this signature feeds a training-free detector that exceeds the strongest internal-feature baseline. Where a hallucination is drawn turns out to carry at least as much information as what it names, and it is accessible to anyone with API access to the model. Because the same features could help adversaries hide hallucinations, we release code under an academic-use licence and add no new images to the public domain; POPE inherits COCO's biases and so does our atlas.

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